

Tutorial 9

Q1. Consider a four-stage pipeline with the stages, Fetch (F), Decode (D), Execute (E), and Writeback (W). The description of the stages is given as follows:

1. Fetch - Fetches the instructions from instruction memory
2. Decode - Decodes the instructions and type of operations. It also reads the input operands. If the input operands are not present, the decode stage is stalled for one cycle to read the operands after they are made available.
 - The address of the unconditional branch is assigned to PC at the end of the decode stage.
3. Execute - Executes the instruction in the ALU
4. Writeback - Updates the register value at the end of the stage.

Considering the above stages, determine the following program's CPI (Cycles Per Instruction).

1. ADD R1, R2, R3
2. SUB R4, R1, R2
3. B 5 // branch to instruction 5
4. SUB R5, R6, R7
5. ADD R4, R5, R6

Q2. Explain the difference between dependency and hazard.

Q3. Consider an **ascending** stack with an initial address of hexadecimal number 0xA00. The address bus is 2 bytes wide. PUSH and POP instructions are used to modify the values of the stack. The data bus is 4 bytes wide. The SP points to the next available memory location in the stack after the instruction is executed. Determine the SP of address for all instructions in the given program.

1. PUSH
2. POP
3. POP
4. POP
5. PUSH
6. POP